How Economists Discuss Central Bank Digital Currencies

Themes, Tone and Stance, 2014–2026

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Abstract

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1 Introduction

2 Related literature

2.1 Central bank digital currencies

2.2 What the field builds on

2.3 Bibliometric and text-as-data studies of CBDCs

Within our corpus, we observe 29 titles that announce a literature, systematic or bibliometric review of CBDCs, published between 2019 and 2026. When comparing the five most recent bibliometric studies with ours, we found three shared main limits. First, their data come from one or two databases, Scopus and/or Web of Science, and are mostly journal papers written in English. Working papers, central-bank speeches and central-bank documents are therefore mostly absent from their analysis, so that their sample is very limited compared to ours: a few hundred documents compared to 4,710. In addition, they build their analysis with standard tools that mainly describe what is studied, but not how (tone and stance) or why. Finally, their classifications are never checked against a reading of the texts themselves.

Our study addresses these three limits. Our corpus is much larger, with papers coming from five databases, central-bank speeches and documents, in different languages, and every text is checked to be written by an economist or an economic institution. We ask not only what is written but also who writes it, how, why and when. Finally, each automatic classification we use is compared with our own reading of a sample of texts.

MISSING : text-as-data studies

3 Data

3.1 Sources and collection

3.2 De-duplication and enrichment

3.3 Relevance: is CBDC the theme?

3.4 Who counts as an economist

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4 Who writes

4.1 Method: co-citation, co-authorship, bibliometric laws and distinctive vocabulary

4.2 Findings

5 What is written

5.1 Method: thematic map, topic model and covariate-adjusted shares

To see what is written by economists, we use two methods. The first one consists in grouping keywords that often appear together in the same papers into themes. Then, we place these themes on a thematic map with two axes: how much a theme is linked to the other themes (*centrality*) and how much its keywords are linked to each other (*density*). This allows us to classify themes into four types: motor, basic, niche, and emerging or declining.

The second method consists in grouping texts by their meaning. To do so, we use *SBERT*, a multilingual model that turns each title and abstract into a vector, so that two texts saying the same thing end

up close, even if they are not written in the same language. We then group these vectors into 24 topics with *BERTopic* and label each topic with its most specific words.

Finally, to make sure that a change in the share of a topic is not only due to a change in who writes, we run a logit regression of each topic on the period, the type of author and the language. That allows us to measure how the share of each topic changed between two periods (here, between 2014-2019 and 2023-2026), in percentage points, all else being equal.

5.2 Findings

The thematic map shows that the CBDC literature is driven by two themes: blockchain technology and the digital transformation of financial services, together with monetary policy and payments. Fintech and digital finance form a basic theme, while digital payments, India and the cashless society are niche topics.

From our analysis, we observe a shift in the way economists discuss CBDC from 2014 to 2026. The early literature treated CBDC as a monetary policy instrument: half of the papers published before 2020 deal with monetary policy and its economic impact, against a fifth today. The topic model shows the same trend, as the topic on interest rates and uncertainty decreases by 9.6 points. Early papers also often discussed CBDC together with private cryptocurrencies: the largest topic of the period is characterized by words such as *cryptocurrencies* and *private*, and it is also the topic that lost the most ground since then (−13.2 points).

After 2020, as central banks started to work on their own projects, the discussion became more national. The digital euro and the digital ruble, almost absent before 2020, now appear in 11% and 15% of the papers, and *euro*, *ruble* and *India* are among the words that best characterize recent years. At the same time, financial inclusion is the theme that grows the fastest, from 6% to 16% of the papers. Overall, the literature moved from general questions on the effects of a CBDC to questions on specific national projects and on who they are meant for.

6 How economists discuss CBDCs: tone and stance

6.1 Method: tone (FinBERT) and stance (reading and a local language model)

6.2 Findings

6.3 Do policy events move the tone?

7 Why: stated motivations

7.1 Method: a validated codebook

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8 When: the policy environment

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9 The discussion and its environment

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A Search vocabulary

A single vocabulary drives both the searches and the relevance score (Section 3). It has 31 concepts, each written as a case-insensitive regular expression, and 115 search phrases derived from them, in 12 languages. Three principles guided it.

1. **Recall first.** The vocabulary covers the generic term in every major language, the names of national projects, the BIS Innovation Hub experiments, and the older names used before "CBDC" became standard (*digital fiat currency*, *central bank cryptocurrency*, *Fedcoin*, *RSCoin*, *CADcoin*, *digital base money*).

2. **Strong and ambiguous terms.** A *strong* term identifies a central bank digital currency on its own (*central bank digital currency*, *digital euro*, *e-CNY*). An *ambiguous* term does not: "CBDC" is also *chronic bullous dermatosis of childhood*, "DCEP" a chemical compound, "Project Stella" a satellite. Ambiguous terms count only when the document also contains monetary context: *currency, monetary, payment, central bank, banking, ledger, stablecoin*, and their equivalents in French, Spanish, Portuguese, German, Italian, Russian, Chinese, Japanese and Korean. When the only evidence is in the body of the text, a stricter context list is required (*central bank, digital currency, monetary, payments*).

3. **Never-scoring terms.** *Digital dollar* and *Sand Dollar* are reported but never scored: they are too often used for private tokens or unrelated products.

Table 1. Concepts of the search vocabulary (selection)

Concept	Forms matched
Generic	CBDC, wCBDC, rCBDC, mCBDC; central bank(s) digital currency/currencies; digital central bank money; tokenised central bank money or reserves
Other languages	monnaie numérique de banque centrale (MNBC); moneda / moeda digital de banco central; digitales Zentralbankgeld; valuta / moneta digitale di banca centrale; Japanese, Korean and Russian equivalents (*tsifrovaya valyuta tsentral'nogo banka*)
Euro area	digital euro, euro numérique, digitaler Euro, euro digitale, euro digital
China	e-CNY, digital yuan / renminbi, DC/EP (Digital Currency Electronic Payment), Chinese forms
Other projects	digital pound (Britcoin), e-krona, digital / e-rupee, eNaira, DCash, JAM-DEX, digital ruble (*tsifrovoy rubl'*), Drex / real digital, e-hryvnia, digital yen, e-HKD, e-peso, Bakong, e-Cedi, digital shekel, lira, won, rial, e-dirham
Early names	digital fiat currency, central bank cryptocurrency, central bank e-money / electronic money / digital cash, sovereign or national digital currency, digital legal tender, Fedcoin, RSCoin, CADcoin
BIS projects	mBridge; Projects Helvetia, Jura, Dunbar, Icebreaker, Aurum, Rosalind, Tourbillon, Mariana, Hamilton, Stella, Ubin, Inthanon, Khokha, Jasper, Agorá, Orchid, Polaris, Sela, Mandala, Ensemble
Adjacent	tokenised deposits, unified ledger (ambiguous: counted only with monetary context)

The full list of regular expressions and search phrases is in `cbdc_biblio/config.py`.

B Harvesting algorithms

B.1 Sources

Records come from four families of sources (Table 2). Nothing is filtered at harvest time: every record found by a search is kept, and relevance is decided afterwards on the merged corpus. Every step writes a checkpoint, so an interrupted harvest resumes where it stopped.

Academic databases. OpenAlex (journal articles, working papers, books, chapters, preprints, reports), Semantic Scholar, arXiv, HAL and DOAJ. Each search phrase is queried in titles and abstracts; results are merged on the source's identifier. OpenAlex reference lists are resolved into citation strings, which the co-citation analysis uses.

Central bankers' speeches. The BIS bulk archive (all central banks, 1996 onwards, full text), the ECB's speech dataset (full text, including non-English originals), and the BIS RSS feed for speeches published after the bulk archive.

Authorities. A crawl starting from 26 CBDC pages of central banks, EU institutions and international organisations, following links one level deep within the same site, PDFs included.

Press. Google News, searched through its RSS interface in 55 combinations of country edition, language and query. GDELT was also queried but throttled the connection for hours after a few requests, so it contributed nothing to the final corpus.

Table 2. Records by source and filtering stage

Source	Collected	CBDC a theme	By economists
Google News	24,309	9,496	0
OpenAlex	20,596	7,266	4,043
Semantic Scholar	12,528	4,000	2,164
DOAJ	1,750	485	251
BIS central bankers' speeches	618	465	465
Central banks and authorities (crawl)	266	239	205
HAL	152	83	31
ECB speeches	127	108	108
arXiv	93	90	24
Distinct records	48,188	17,709	4,710

A record found in several sources counts once per source; the last row counts distinct records.

B.2 Pagination and rate limits

Academic APIs return results page by page; the harvester follows the API's cursor (200 records per page for OpenAlex) until no page is left. Requests are spaced by a minimum interval Δ,

$$t_{k+1} - t_k \geq \Delta,$$

with $\Delta = 0.12$ s for OpenAlex, 3.5 s for arXiv and 1.2 s for Google News. After the j-th consecutive refusal (HTTP 429 or 5xx), the harvester waits $w_j = \min(60j, 900)$ seconds before retrying.

B.3 Press: recursive bisection of time windows

A Google News search returns at most about 100 items, whereas a single month can hold thousands of articles. Each search is therefore run on time windows that are split until every window falls below the cap:

1. Start from monthly windows $W = [a, b)$ covering January 2000 to the harvest date, on a stack.
2. Pop a window and fetch the items published in it.
3. If it returns $N(W) \geq 95$ items and lasts more than one day, split it at $m = a + (b - a)/2$, push $[m, b)$ and $[a, m)$, and go back to step 2.

4. Otherwise save the items and checkpoint the window.

A month is split at most $\lceil \log_2 31 \rceil = 5$ times, and keeping L windows costs $2L - 1$ requests. Google News limits requests over time, and the harvest was stopped with about 15 of the 55 searches unfinished; press counts are therefore incomplete, and skewed towards recent years.

B.4 Full text, enrichment and de-duplication

Full text. PDFs are extracted with PyMuPDF and web pages with trafilatura, which also recovers the page's date, author and outlet when the record lacks them. Texts are split into introduction, conclusion and section headings; for speeches and press, the opening and closing paragraphs play these roles.

Enrichment. Language is detected with `langdetect`; keywords are extracted with YAKE (one to three words) when a record has no author or indexed keywords. Dates are kept with their precision (day, month, year), and 1 January dates are flagged, since they often stand for a year only.

De-duplication. Records are matched on a key: the DOI for academic records, else their normalised title and year; the date and normalised title for speeches; the URL for other records. A second pass merges academic records with the same normalised title and year but different DOIs (a preprint and its published version, for instance). Merged records keep the most precise date, the longest abstract and the union of keywords and sources.

B.5 Relevance score

Each available field f of a record receives a centrality $c_f \in [0, 1]$: 1 for a title containing a strong term; 0.4 for a keyword list containing one (keywords are partly machine-assigned); for the abstract, the share of its sentences that contain one; for the body, half the density of mentions (per 1,000 words, saturating at 3) and half their spread (the share of ten equal slices of the text that contain one). The score is

$$\text{score} = 100\Big(0.6 \max_{f \in F} c_f + 0.4 \frac{1}{|F|} \sum_{f \in F} c_f\Big),$$

where F is the set of fields available, so a missing abstract or full text is not penalised. Records scoring at least 50 are labelled *main theme*, from 25 to 49 *secondary theme*, and below 25 *mentioned only*. Records whose only matches are ambiguous terms without monetary context are *ambiguous terms only*; records without any match are *off-topic*; press items whose text could not be retrieved are *unverified*. Records with CBDC as a theme but dated before 2014 were all homonyms or misdated, and are labelled *dated before 2014*. Table 3 gives the labels and their validation.

Table 3. Relevance labels and their validation

Label	Records	Read	Theme	95% CI
main theme	15,464	30	29	0.83–0.99
secondary theme	2,245	29	11	0.23–0.56
mentioned only	489	30	1	0.01–0.17
unverified (no text)	13,446	19	2	0.03–0.31
ambiguous terms only	1,709	10	3	0.11–0.60
off-topic	14,803	20	0	0.00–0.16
dated before 2014	32	0	0	–

Read: records of the stratified sample (138 texts) read blind to the label; Theme: read as having CBDC as a theme; Wilson intervals on the share.

B.6 Who counts as an economist

For each academic record with CBDC as a theme, the OpenAlex field of its primary topic and the institutions of its authors were retrieved (`scripts/tag_economists.py`). A text is attributed to economists when it is research classified in *Economics, Econometrics and Finance*, or research with an author at a central bank, the IMF or the BIS, or, for records absent from OpenAlex, research published in an economics venue; when it is a central bank speech; or when it is a document of a central bank or an international financial institution. Tables 4 and 5 show the result.

Table 4. Who counts as an economist

Texts with CBDC as a theme	Texts	
Research classified in Economics, Econometrics and Finance	3,783	kept
Research in another field, with a central bank, IMF or BIS author	148	kept
Research without OpenAlex record, in an economics venue	63	kept
Research in other fields	3,469	dropped
Central bank speeches	511	kept
Documents of central banks and international institutions	205	kept
Documents of governments and parliaments	34	dropped
Press articles	9,496	dropped
Texts by economists	4,710	

Field: OpenAlex primary topic of the work; author institutions from OpenAlex and from the record's affiliations.

Table 5. Research with CBDC as a theme left out by the economist filter, by field

OpenAlex field of research dropped	Texts
Computer Science	1,769
Business, Management and Accounting	663
no OpenAlex record	473
Social Sciences	281
Decision Sciences	244
Engineering	14
Medicine	7
Health Professions	5
Environmental Science	4
Arts and Humanities	3
Agricultural and Biological Sciences	2
Neuroscience	2
Immunology and Microbiology	1
Materials Science	1

C Validation samples and coding guide

Every measure used in the paper was compared with a reading of the texts. The samples were drawn at random; the motivation, stance and tone samples were topped up with texts written by economists so that each holds at least 50 of them (Table 6). Coding was done blind to the machine labels, by one coder assisted by a language model; a 200-item sheet for a second, independent human coder is provided with the data (`presentation/second_coder_sheet.csv`).

C.1 Coding guide

Relevance. Is a central bank digital currency the theme of the text, or one of its main themes? A passing mention, a homonym, or a text about cryptocurrencies or stablecoins only is coded *no*.

Table 6. Hand-coded validation samples

Sample	Units	From economists' texts
Relevance	140	25
Motivations, development	100	60
Motivations, test (held out)	74	50
Stance	177	100
Tone (sentences)	136	90

Motivations. Unit: the first 1,500 characters of the introduction, else of the abstract, else of the text. A motivation is coded when the text raises it as a reason to study, issue or worry about a CBDC; a text can have several. The ten motivations are defined as follows.

- *Decline of cash*: cash use is falling, or access to cash or public money is at risk.
- *Financial inclusion*: reaching unbanked or underbanked people; access to financial services.
- *Monetary sovereignty*: a threat from private money, stablecoins, Big Tech, cryptocurrencies or foreign currencies to the central bank's control of money.
- *Payment efficiency*: cheaper, faster, more efficient, resilient or innovative domestic payments.
- *Cross-border payments*: international payments, remittances, cross-border settlement.
- *Monetary policy*: monetary policy, interest rates, transmission, money supply.
- *Financial stability*: bank disintermediation, bank runs, deposit outflows, risks to financial stability.
- *Privacy and surveillance*: privacy, anonymity, data protection, surveillance or control of users.
- *Geopolitics and sanctions*: geopolitics, sanctions, dollar dominance, internationalisation of a currency.
- *Technology*: blockchain, distributed ledgers, tokenisation, smart contracts, programmable money.

Stance. Unit: the title and abstract (research), or the title and the first five sentences that name a CBDC (speeches, documents). *Favourable*: supports CBDCs or presents them positively; *critical*: opposes them or stresses their risks and problems; *neutral*: descriptive or balanced. Analysing risks is neutral; arguing that they can be solved so that a CBDC can be issued, or recommending issuance, is favourable.

Tone. Unit: one English sentence that names a CBDC. *Positive* or *negative* when the sentence presents CBDCs or their effects favourably or unfavourably; *neutral* when it describes, defines or reports.

C.2 Results

The motivation codebook was written on the 100 development texts only and evaluated on the held-out test texts (Table 7). Each alternative coder was tested on the same texts (Tables 8 and 9); thresholds and classifiers were set on the development texts (motivations) or by 5-fold cross-validation (stance). Agreement is measured by Cohen's $\kappa = (p_o - p_e)/(1 - p_e)$, where p_o is the observed and p_e the chance agreement. The language model's stance labels are used only for changes over time: it labels too many texts favourable, a bias that cancels in a difference if it is stable. Stance levels come from reading (Table 11).

Table 7. Motivation codebook on the held-out texts written by economists

Motivation	Read	Coded	Precision	Recall	κ
Decline of cash	7	1	1.00	0.14	0.22
Financial inclusion	7	3	1.00	0.43	0.56
Monetary sovereignty	6	1	1.00	0.17	0.26
Payment efficiency	16	3	0.67	0.12	0.12
Cross-border payments	1	3	0.33	1.00	0.48
Monetary policy	7	9	0.56	0.71	0.55
Financial stability	10	7	0.57	0.40	0.37
Privacy and surveillance	3	2	1.00	0.67	0.79
Geopolitics and sanctions	7	4	1.00	0.57	0.70
Technology (DLT, programmability)	11	11	0.73	0.73	0.65
All (pooled)	75	44	0.70	0.41	0.46

50 held-out texts by economists (of 74), never used to write the codebook. Read: texts where reading found the motivation; Coded: texts flagged by the codebook.

Table 8. Candidate coders for motivations

Coder	P	R	F_1	κ	κ, all
Codebook (regex)	0.70	0.41	0.52	0.46	0.50
Codebook OR classifier	0.42	0.60	0.50	0.39	0.42
SBERT, classifier (100 labels)	0.36	0.37	0.37	0.25	0.28
Local LLM, Qwen2.5 7B	0.30	0.61	0.40	0.25	0.28
Local LLM, Qwen2.5 3B	0.31	0.23	0.26	0.15	0.16
SBERT, definitions (zero-shot)	0.21	0.71	0.32	0.12	0.15
NLI zero-shot (mDeBERTa)	0.22	0.39	0.28	0.11	0.14

Scores on the 50 texts written by economists; last column: κ on all 74 hand-coded texts. Thresholds and classifiers set on the 100 development texts.

Table 9. Candidate coders for stance

Coder	Accuracy	κ	κ, all
Local LLM, Qwen2.5 7B	0.70	0.45	0.41
FinBERT tone	0.65	0.31	0.20
NLI zero-shot (mDeBERTa)	0.63	0.29	0.18
SBERT classifier	0.61	0.21	0.17

Scores on the 100 texts written by economists; last column: κ on all 177 hand-coded texts. Thresholds and classifiers by 5-fold cross-validation.

Table 10. FinBERT tone against reading

FinBERT ↓ / reading →	negative	neutral	positive	Precision
negative	17	11	2	0.57
neutral	2	25	3	0.83
positive	1	10	19	0.63

90 sentences from economists' texts, 30 per FinBERT label, read blind to the label: agreement 68%, $\kappa = 0.52$.

Table 11. Stance by reading, texts written by economists

Actor	n	Favourable	Neutral	Critical
Research economists	50	8 [3–19]	90 [79–96]	2 [0–10]
Central banks and institutions	50	52 [39–65]	42 [29–56]	6 [2–16]

Percentages with Wilson 95% intervals.

D Additional tables and figures

Table 12 lists the policy events used in the event study and the test of tone (Sections 6 and 8), with the attention each one received. Table 13 lists every topic found by the topic model (Section 5).

Table 12. Policy events and the attention they received

Date	Event	Texts	Expected	Ratio	p
18 Jun 2019	Facebook announces Libra	1	3.4	0.3	0.83
02 Oct 2020	ECB digital euro report	13	4.5	2.9	0.01
20 Oct 2020	Bahamas launches Sand Dollar	17	5.7	3.0	0.01
14 Jul 2021	ECB investigation phase	6	13.0	0.5	0.93
25 Oct 2021	Nigeria launches eNaira	9	11.4	0.8	0.69
20 Jan 2022	Fed CBDC discussion paper	14	10.1	1.4	0.24
09 Mar 2022	US order on digital assets	6	10.3	0.6	0.84
01 Nov 2022	India e-rupee pilot	14	14.6	1.0	0.55
07 Feb 2023	UK digital pound consultation	10	13.8	0.7	0.76
28 Jun 2023	EU digital euro bill	32	19.3	1.7	0.05
18 Oct 2023	ECB preparation phase	63	19.1	3.3	0.00
23 Jan 2025	US order banning a US CBDC	26	21.9	1.2	0.29

Texts by economists dated to the day, in the 15 days from the event; expected from the daily rate over days −90 to −15. p: share of 500 random dates with a larger $z = (N - E)/\sqrt{E}$.

Table 13. Topics of the economists' corpus

	Topic (top terms)	Texts	2014–2019	2020–2022	2023–2026
1	cryptocurrencies, implementation, private, traditional, national	470	18.5	13.5	8.5
2	design, retail, legal, private, privacy	439	10.4	11.2	8.6
3	uncertainty, rates, design, rate, interest	349	14.2	8.7	6.7
4	dollar, trade, debt, reserve, inflation	326	5.7	5.8	7.5
5	deposits, interest, deposit, rate, effects	319	2.8	5.0	7.3
6	governor, innovation, fintech, kong, hong	306	17.5	11.2	4.7
7	russian, russia, ruble, federation, legal	238	0.0	5.1	4.9
8	china, rmb, e-cny, yuan, internationalization	213	1.4	5.3	4.6
9	cashless, evolution, riksbank, forms, physical	204	6.6	4.9	4.1
10	fintech, technologies, services, innovation, finance	199	7.1	2.9	4.4
11	euro, european, ecb, area, legal	186	0.5	3.6	4.2
12	cryptocurrencies, crypto, bitcoin, cryptocurrency, assets	186	9.0	4.0	3.6
13	journal, list, approved, chapter, implementation	176	0.5	3.4	4.2
14	india, rupee, indian, rbi, upi	132	0.0	1.1	3.5
15	ruble, russia, russian, form, platform	130	0.0	1.9	3.2
16	cross-border, interoperability, settlement, instant, mbridge	127	0.9	2.1	2.9
17	euro, european, europe, ecb, parliament	127	0.9	2.4	3.0
18	euro, sie, european, auswahl, textabschnitt	110	0.5	1.3	2.8
19	nigeria, enaira, african, africa, inclusion	103	0.5	1.6	2.4
20	legal, assets, virtual, law, regulation	101	0.9	1.5	2.3
21	euro, project, rulebook, latest, in-depth information	77	0.5	0.6	2.1
22	raqamli, markaziy, circular, tahlil, joriy	73	0.0	0.7	1.6
23	ukraine, e-hryvnia, ukrainian, electronic, civil	69	1.4	1.3	1.6
24	ruble, legal, civil, agreement, platform	50	0.0	0.6	1.3

BERTopic on 4,710 texts (multilingual SBERT embeddings, UMAP, HDBSCAN with leaf selection and at least 40 texts per topic); 50% of texts were outliers, assigned to the nearest topic. Terms: class-based TF-IDF on English texts. Period columns: % of the period's texts.

E Data and code availability

E.1 Data

The corpus is a single file, `data/cbdc_corpus.csv`: one row per document and 49 columns, in UTF-8. Every record keeps its issuer, its date with the precision of that date, its original language, its keywords (author, indexed and extracted), its abstract, introduction and conclusion when available, its full text, its relevance label and theme score, and the sources it was found in. The harvest ran from 8 to 11 October 2026; it covers 48,188 documents, of which 17,709 have CBDC as a theme and 4,710 were written by economists (Section B). The build step can also export it in two formats read by bibliometric software: a Scopus-style CSV (for `bibliometrix` and VOSviewer) and RIS.

Caches hold the results that are slow to recompute: sentence embeddings, sentence tones, the language model's stance labels, and the OpenAlex fields and institutions of each paper. The hand-coded validation samples and every coder's predictions are in `presentation/` (Table 6).

Many full texts are under copyright. The data are therefore shared with the members of the project and on request, not published openly. The metadata, identifiers (DOI, OpenAlex and source identifiers, URLs) and the code that rebuilds the corpus from the public sources are not restricted.

E.2 Code

All code is in Python and kept in a version-controlled repository, available from the authors on request. One command per step rebuilds the corpus, and every step resumes where it stopped:

1. `run.py harvest`: query every source (Section B);
2. `run.py fulltext`: download and extract the full texts;
3. `run.py build`: merge, de-duplicate, enrich and score relevance;
4. `scripts/tag_economists.py`: retrieve the OpenAlex field and institutions of each paper;
5. `scripts/validate_coders.py`: score every candidate coder against the hand-coded samples;
6. `scripts/make_appendix.py`: write the tables of this appendix (`paper/tables/`).

The analyses (networks, topics, motivations, tone, stance, event studies, robustness) are in `cbdc_biblio/analysis.py` and `cbdc_biblio/nlp.py`.

E.3 Software and models

Python 3.10 or later; the packages and their versions are listed in `requirements.txt`. The models are open and run locally:

- sentence embeddings: `paraphrase-multilingual-MiniLM-L12-v2` (Sentence-Transformers);
- tone: FinBERT (`ProsusAI/finbert`);
- natural language inference (tested coder): `mDeBERTa-v3-base-xnli-multilingual-nli-2mil7`;
- stance: Qwen2.5 7B Instruct, 4-bit, run with `mlx-lm` (Apple Silicon); Qwen2.5 3B was tested.

Everything ran on a laptop (Apple M1, 8 GB of memory). Coding new texts with the language model requires Apple Silicon; every other step runs on any machine once the caches are present.